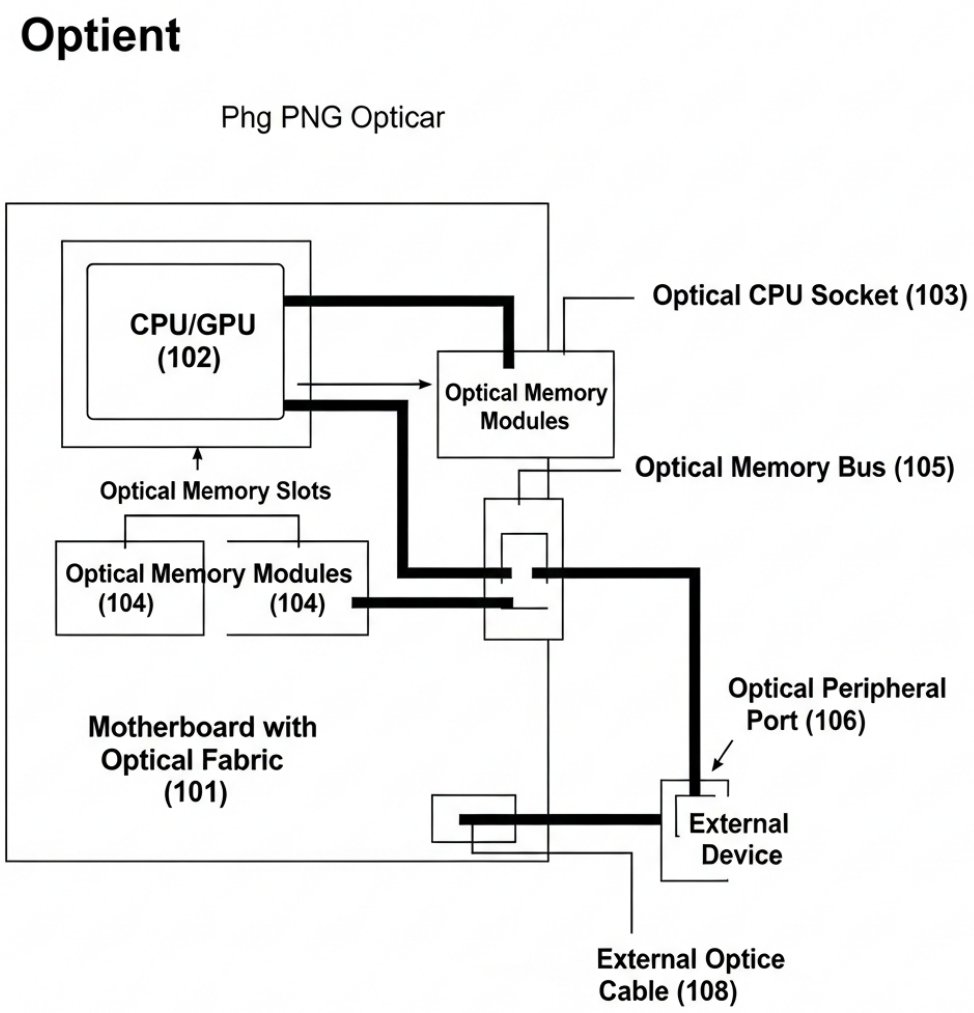
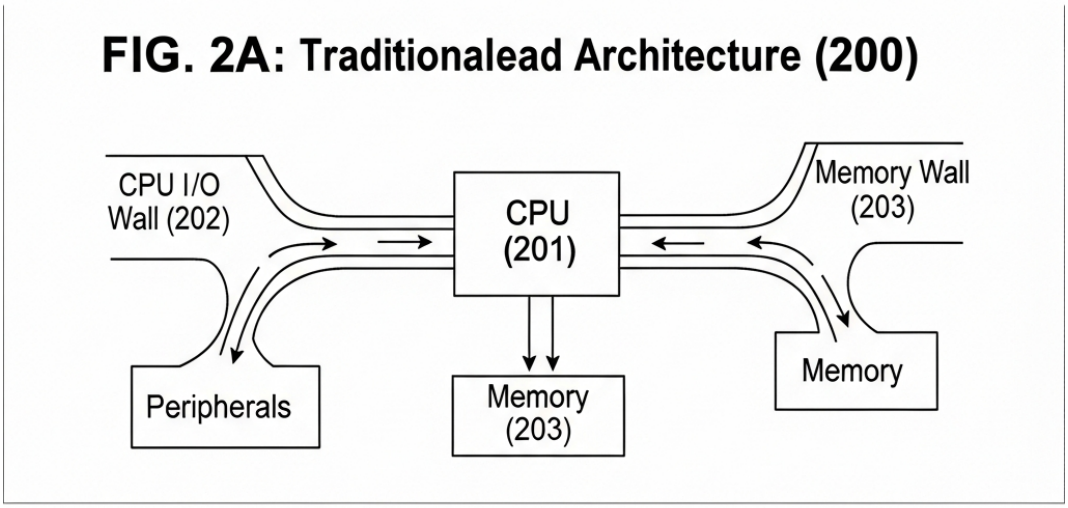

Optient
Phg PNG Opticar
CPU/GPU
(102)
Optical CPU Socket (103)
Optical Memory
Modules
Optical Memory Bus (105)
Optical Memory Slots
Optical Memory Modules
(104)
(104)
Optical Peripheral
Port (106)
Motherboard with
Optical Fabric
(101)
External
Device
External Optice
Cable (108)

FIG. 1


FIG. 2A: Traditionalead Architecture (200)
CPU I/O Wall (202)
CPU (201)
Memory Wall (203)
Peripherals
Memory (203)
Memory

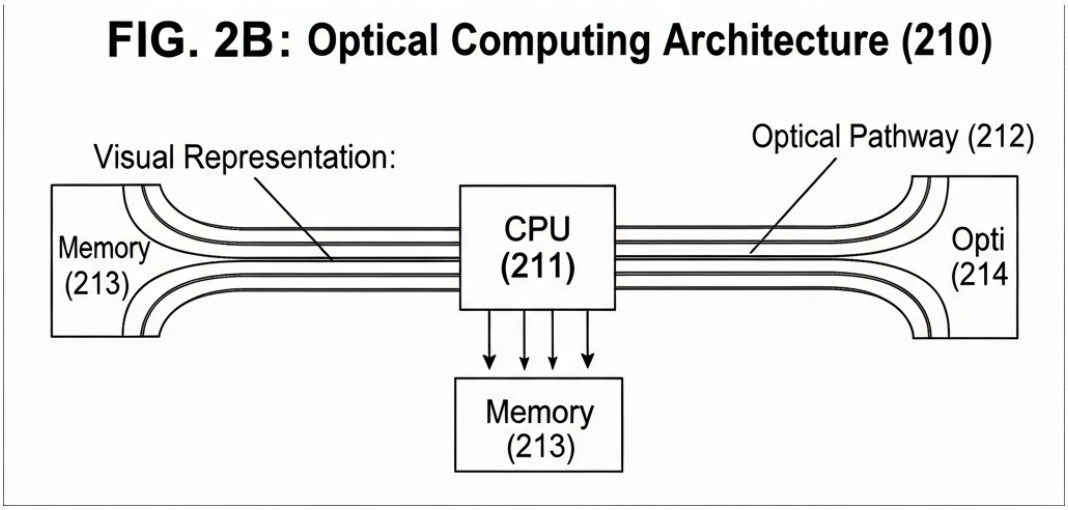

FIG. 2B: Optical Computing Architecture (210)
Visual Representation:
Optical Pathway (212)
Memory (213)
CPU (211)
Opti (214
Memory (213)

FIG. 2

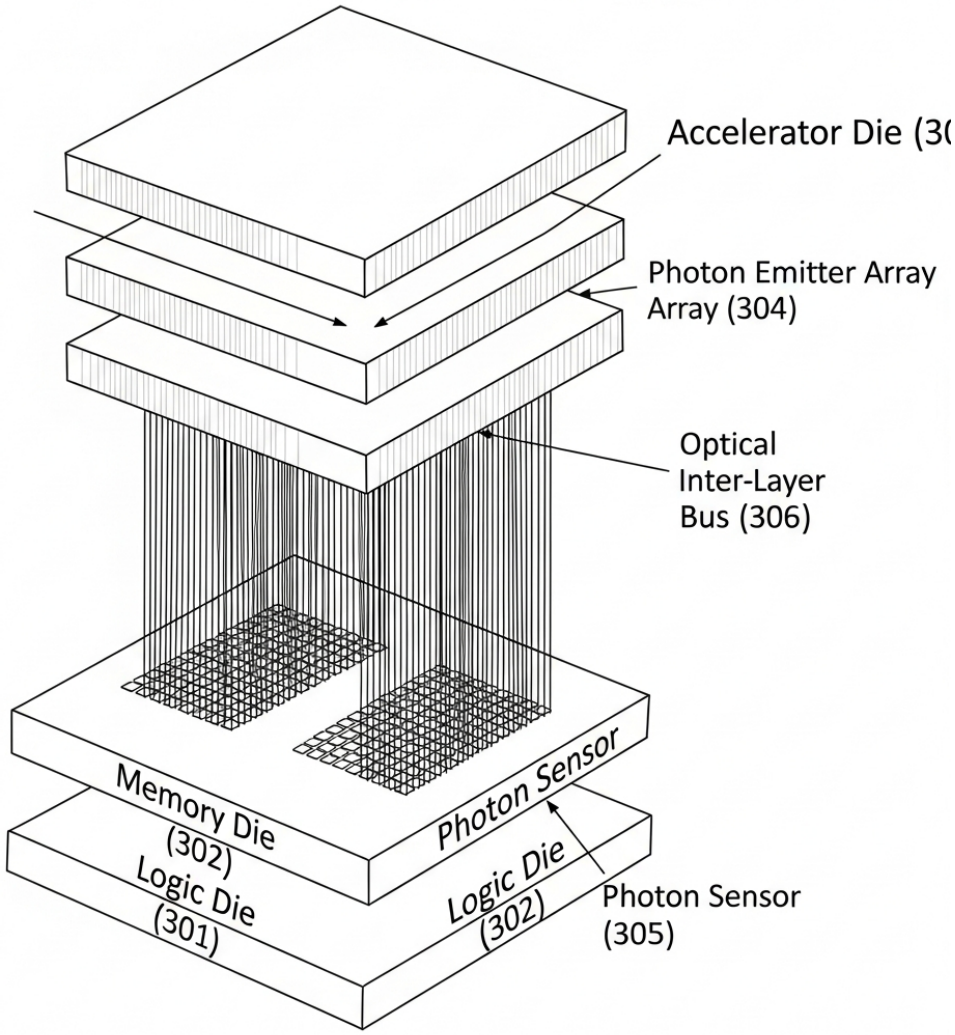

Accelerator Die (3(
Photon Emitter Array
Array (304)
Optical
Inter-Layer
Bus (306)
Memory Die
(302)
Photon Sensor
Logic Die
(301)
Logic Die
(302)
Photon Sensor
(305)

FIG. 3